ON SEPARATING SEMIGROUPS OF $(M-2)$-CURVES

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Abstract. We study the separating semigroups of separating $(M-2)$-curves, that is, real algebraic curves of genus g whose real loci have $g-1$ connected components.

For $g \geqslant 5$, we prove that $\mathrm{Sep}(C) = \mathbb{N}^{g-1}$ for every separating $(M-2)$-curve C not of special type in the sense of Coppens. Consequently, the equality $\mathrm{Sep}(C) = \mathbb{N}^{g-1}$ holds on a non-empty open subset of the Teichmüller space of separating $(M-2)$-curves.

Along the way, we prove that a separating $(M-2)$-curve C of genus $g \geqslant 5$ with separating gonality $g-1$ is of special type if and only if every separating pencil of degree $g-1$ on C is a theta characteristic.

For curves of special type, we obtain partial descriptions of $\mathrm{Sep}(C)$ that generalize the known low-genus results.

§1. Introduction

By a *real algebraic curve* C we mean a complex algebraic curve endowed with an antiholomorphic involution $\mathrm{conj}\colon C \to C$, called a *real structure*. The set of real points of C, denoted by $\mathbb{R}C$, is the fixed-point set of conj. All curves considered here are projective, smooth, and irreducible.

A real curve C is called *separating* or *dividing* if $C \setminus \mathbb{R}C$ is disconnected. In this case $C \setminus \mathbb{R}C$ consists of two connected components which are interchanged by an antiholomorphic involution. The boundary orientation induced on $\mathbb{R}C$ by one of the halves of $C \setminus \mathbb{R}C$ is called *a complex orientation*.

Due to a classical result of Ahlfors [Ahl50], a real curve C is separating if and only if it admits a *separating morphism*, that is, a real morphism $f\colon C \to \mathbb{P}^1$ such that $f^{-1}(\mathbb{RP}^1) = \mathbb{R}C$. Therefore, the study of separating curves is closely related to the study of separating morphisms.

Every separating morphism defines a covering map $f|_{\mathbb{R}C}\colon \mathbb{R}C \to \mathbb{RP}^1$. Fix a numbering of the connected components $C_1, \dots, C_r$ of $\mathbb{R}C$. Denoting by $d_i(f)$ the degree of the restriction of f to C_i, we can assign to each separating morphism f the tuple of positive integers $d(f) = (d_1(f), \dots, d_r(f))$. Kummer and Shaw defined the *separating semigroup* of C as the set of all such tuples:

$$\mathrm{Sep}(C) := \{d(f) \mid f\colon C \to \mathbb{P}^1 \text{ is a separating morphism}\}.$$

It is shown in [KS20, Prop. 2.1] that $\mathrm{Sep}(C)$ is indeed a semigroup. We set

$$\mathbb{N} := \{n \in \mathbb{Z} \mid n \geqslant 1\}, \qquad \mathbb{N}_0 := \{n \in \mathbb{Z} \mid n \geqslant 0\}.$$

By Harnack's inequality, we have $b_0(\mathbb{R}C) \leqslant g+1$ for every real curve C of genus g. A real curve C is called an *M-curve* if $b_0(\mathbb{R}C) = g+1$. If $b_0(\mathbb{R}C) = g+1-i$, then C is called an *$(M-i)$-curve*. Kummer and Shaw [KS20, Theorem 1.7] proved that for an M-curve C of genus g we have $\mathrm{Sep}(C) = \mathbb{N}^{g+1}$. By Klein's congruence, for a separating $(M-i)$-curve C we have $b_0(\mathbb{R}C) \equiv g+1 \pmod 2$, so i must be even. Consequently, after M-curves, the next natural case to consider is that of *$(M-2)$-curves*. The present paper is devoted to describing their separating semigroups.

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The semigroup $\mathrm{Sep}(C)$ has already been computed for separating $(M-2)$-curves of genus 3 [Ore20] and 4 [Ore25]. For separating plane $(M-2)$-quintics, which have genus 6, $\mathrm{Sep}(C)$ is computed in [MO25]. These results are listed in Table 1.

The description of the separating semigroup depends crucially on the minimal degree of a separating morphism. We denote this value by $\mathrm{sepgon}(C)$ and call it the *separating gonality* of C. It follows from results of A. Gabard [Gab06] that the separating gonality of a separating $(M-2)$-curve of genus g is either $g-1$ or g.

In [Man26], M. Manzaroli studies the relation between separating morphisms from separating plane $(M-2)$-curves of degree d to $\mathbb{P}^1$ and totally real pencils of plane curves of degree $d-3$. That paper also contains partial results on the separating semigroups of $(M-2)$-curves of arbitrary genus. More precisely, it is shown that $(3,\dots,3)+\mathbb{N}_0^{g-1}\subseteq\mathrm{Sep}(C)$ if $\mathrm{sepgon}(C)=g-1$, and $(4,2,\dots,2)+\mathbb{N}_0^{g-1}\subseteq\mathrm{Sep}(C)$ if $\mathrm{sepgon}(C)=g$.

Besides separating gonality, an important role is played by the notion of *special type* introduced by Coppens [Cop14]. A curve C is *of special type* if $\mathbb{R}C$ has a component such that every real morphism of degree g with odd topological degree on all other real components must be separating. We recall some basic properties of this notion in §2.2.

Main results. For curves not of special type, we obtain a complete answer: as in the case of M-curves, all degree partitions are realizable.

Theorem A (Theorem 3.7)**.** *Let C be a separating $(M-2)$-curve of genus $g\geqslant 5$ that is not of special type. Then*

$$\mathrm{Sep}(C)=\mathbb{N}^{g-1}.$$

Coppens [Cop14, Lemma 2.6 and Corollary 4.2] proved that curves not of special type form a non-empty open subset of the Teichmüller space of separating $(M-2)$-curves of genus $g\geqslant 5$. By Theorem A, every curve C in this subset satisfies $\mathrm{Sep}(C)=\mathbb{N}^{g-1}$ (Corollary 3.8).

It turns out that, for separating $(M-2)$-curves, separating gonality and special type are closely related to theta characteristics. A key step in the proof of Theorem A is to show that every separating $(M-2)$-curve of genus $g\geqslant 5$ that is not of special type admits a separating pencil of degree $g-1$ that is not a theta characteristic (Theorem 3.5). Coppens [Cop14] proved that every separating pencil of degree $g-1$ on a curve of special type is a theta characteristic. Combining these results, we obtain a characterization of special type among curves with separating gonality $g-1$: for $g\geqslant 5$, C is of special type if and only if every separating pencil of degree $g-1$ on C is a theta characteristic (Corollary 3.6).

We now turn to curves of special type. In this case we obtain the following inclusions.

Theorem B (Theorem 3.1)**.** *Let C be a separating $(M-2)$-curve of genus $g\geqslant 4$.*

(a) *Assume that $\mathrm{sepgon}(C)=g$ and let f be a separating morphism of degree g. Number the connected components of $\mathbb{R}C$ so that $d_1(f)=2$. Then*

$$(2,1,\dots,1)+\mathbb{N}_0^{g-1}\subseteq\mathrm{Sep}(C).$$

(b) *Assume that $\mathrm{sepgon}(C)=g-1$ and C is of special type with a special component C_1. Then*

$$\{(1,\dots,1)\}\cup\left((2,1,\dots,1)+\mathbb{N}_0^{g-1}\right)\subseteq\mathrm{Sep}(C).$$

For $g=4$, both inclusions are equalities [Ore25]. Equality in part (a) is also known for separating plane $(M-2)$-quintics (which have genus 6); see [MO25]. This leads to the following conjecture.

Conjecture 1.1. *The inclusions in Theorem B are equalities.*

Even for separating plane $(M-2)$-quintics, proving the reverse inclusion requires a delicate argument that relies heavily on the fact that the curve is plane; see [MO25, Lemma 4.3].

Organization of the paper. The paper is organized as follows. In §2, we recall some facts about separating divisors and the theorem of Mikhalkin and Orevkov on the extendability of separating divisors of degrees g and $g-1$. We then discuss special type and its relation to separating gonality, recalling the results of Coppens. Finally, we recall Martens' theorem and prove a dimension estimate needed in the proofs of the main results.

In §3, we prove the main results. We first prove the inclusions of Theorem 3.1 for curves of special type. We then turn to curves not of special type and prove the existence of a separating pencil of degree $g-1$ which is not a theta characteristic (Theorem 3.5). Applying the theorem of Mikhalkin and Orevkov then gives Theorem 3.7.

§2. Preliminaries

Throughout this section, C denotes a smooth projective real algebraic curve of genus g (not necessarily separating). We assume that C has real points, i.e. $\mathbb{R}C \neq \varnothing$. We denote the number of connected components of $\mathbb{R}C$ by l.

As usual, for a divisor D on C, we write $h^i(D) := \dim H^i(C, \mathcal{O}_C(D))$. If $|D|$ is a base-point-free complete linear system of rank r, we denote by $f_{|D|}\colon C \to \mathbb{P}^r$ the associated morphism. This morphism is defined only up to postcomposition with an automorphism of $\mathbb{P}^r$, or equivalently, up to a change of basis in $H^0(C, \mathcal{O}_C(D))$.

A divisor D is called *real* if $\operatorname{conj}(D) = D$. A real effective divisor D is called *totally real* if $\operatorname{Supp}(D) \subset \mathbb{R}C$.

The involution conj on C induces an antiholomorphic involution on $\operatorname{Pic}^d(C)$, given by $[D] \mapsto [\operatorname{conj}(D)]$. We denote its fixed-point set by $\mathbb{R}\operatorname{Pic}^d(C)$. The space $\mathbb{R}\operatorname{Pic}^d(C)$ is diffeomorphic to a disjoint union of 2^{l-1} real tori of dimension g.

A divisor class $[D] \in \operatorname{Pic}^{g-1}(C)$ is called a *theta characteristic* if $2D \sim K_C$. By definition, any two theta characteristics differ by an element of the 2-torsion subgroup of $\operatorname{Pic}^0(C)$. Since this subgroup is isomorphic to $(\mathbb{Z}/2\mathbb{Z})^{2g}$ (see e.g. [GH81, Proposition 1.1]), there are exactly 2^{2g} theta characteristics on C.

The following elementary observation about totally real divisors of small degree will be useful for us.

Proposition 2.1 ([Hui01, Corollary 2.2])**.** *Let $k < l$ be a positive integer, and let $p_1, \dots, p_k \in \mathbb{R}C$ lie on distinct connected components of $\mathbb{R}C$. Then the divisor $D = p_1 + \dots + p_k$ is rigid, that is, $h^0(D) = 1$.*

§2.1. Separating divisors.

We now assume that C is separating and denote the connected components of $\mathbb{R}C$ by $C_1, \dots, C_l$.

Definition 2.2. A *separating divisor* on C is the divisor of zeros of a separating morphism. For a separating divisor $D = (f)_0$ we denote its degree partition vector by $d(D) := d(f) \in \mathbb{N}^l$.

The following proposition describes some properties of separating divisors of degree l.

Proposition 2.3. *Let D be a divisor on C of degree l.*

(a) *If D is separating, then $|D|$ is a pencil, that is, $h^0(D) = 2$.*
(b) *Suppose that $D = p_1 + \dots + p_l$ with $p_i \in C_i$. If $h^0(D) \geqslant 2$ then D is separating.*

Proof. (a) Since $\deg D = l$ and a fiber of a separating morphism meets every connected component of $\mathbb{R}C$, we can write $D = p_1 + \dots + p_l$ with $p_i \in C_i$ for all $i = 1, \dots, l$. By Proposition 2.1, we have $h^0(D - p_i) = 1$, and hence $h^0(D) \leqslant h^0(D - p_i) + 1 = 2$. On the other hand, since D is a fiber of a separating morphism, we have $h^0(D) \geqslant 2$.

(b) By Proposition 2.1 we have $h^0(D-p_i) = 1 < h^0(D)$, so $|D|$ is base-point-free. Hence there exists a real divisor $E \in |D|$ such that

$$\operatorname{Supp}(D) \cap \operatorname{Supp}(E) = \varnothing.$$

As $E \sim D$, we have $\deg(E|_{C_i}) \equiv \deg(D|_{C_i}) \pmod 2$ (see e.g. [Hui03, Corollary 2.4]). Thus E has odd degree on every connected component of $\mathbb{R}C$. Since $\deg E = l$, it follows that E has exactly one point on each connected component of $\mathbb{R}C$. Hence the divisors D and E interlace: every connected component of $\mathbb{R}C \setminus E$ contains exactly one point of D, and vice versa. It follows from [KS20, Lemma 2.10] that D is separating.

□

Following [MO26], we call an element $d \in \operatorname{Sep}(C)$ *extendable* if

$$d + \mathbb{N}_0^l \subseteq \operatorname{Sep}(C).$$

We also call a separating divisor P on C *extendable* if $d(P)$ is extendable. Kummer and Shaw [KS20, Proposition 3.2] proved that every non-special separating divisor is extendable.

As noted above, Gabard's bound [Gab06] implies that every separating $(M-2)$-curve of genus g admits a separating divisor of degree either $g-1$ or g. Fortunately, the following recent theorem of Mikhalkin and Orevkov gives sufficient conditions for separating divisors of these degrees to be extendable.

Theorem 2.4 ([MO26])**.** *Let C be a separating real curve of genus g.*

(a) *Let $P = p_1 + \ldots + p_g$ be a separating divisor on C with $h^0(P) = 2$. Then P is extendable.*

(b) *Assume that C is not hyperelliptic. Let $Q = q_1 + \ldots + q_{g-1}$ be a separating divisor with $h^0(Q) = 2$ such that Q is not a theta characteristic. Then Q is extendable.*

§2.2. Special type and separating gonality. In this subsection, C denotes a separating $(M-2)$-curve. Denote by Teich_g the Teichmüller space of such curves.

Gabard's bound [Gab06] implies that the separating gonality of C is either $g-1$ or g. This gives the following partition of the Teichmüller space:

$$\operatorname{Teich}_g = \operatorname{Teich}_{g,g} \cup \operatorname{Teich}_{g,g-1}. \tag{†}$$

M. Coppens [Cop14] introduced another partition of Teich_g, defined in terms of curves of *special type*.

Definition 2.5. We say that a separating $(M-2)$-curve C of genus g is *of special type* if $\mathbb{R}C$ has a connected component C' with the following property: for every real morphism $f\colon C \to \mathbb{P}^1$ of degree g whose restriction to every other connected component of $\mathbb{R}C$ has odd topological degree, we have $f(C') = \mathbb{RP}^1$. Such a component C' is called *special*.

Otherwise, C is said to be *not of special type*.

For a real morphism $f\colon C \to \mathbb{P}^1$, we define its *topological degree vector* by

$$\delta(f) = \big(\delta_1(f), \ldots, \delta_{g-1}(f)\big) \in \mathbb{Z}^l,$$

where $\delta_i(f)$ is the topological degree of the map

$$f|_{C_i}\colon C_i \approx S^1 \longrightarrow \mathbb{RP}^1 \approx S^1.$$

If f is separating, then $\delta(f)$ is simply the degree of the oriented covering $\mathbb{R}C \to \mathbb{RP}^1$, hence $d(f) = \pm\,\delta(f)$.

Remark 2.6. A real morphism $C \to \mathbb{P}^1$ associated with a complete base-point-free pencil $|D|$ is determined only up to postcomposition with a real automorphism of $\mathbb{P}^1$, or equivalently, up to a change of real basis in $H^0(C, \mathcal{O}_C(D))$. Since such an automorphism may reverse the orientation of $\mathbb{RP}^1$, the vector $\delta(f_{|D|})$ is determined only up to a simultaneous change of sign in all its components.

Remark 2.7. Definition 2.5 can be reformulated as follows: a component $C' \subset \mathbb{R}C$ is special if every morphism $f\colon C \to \mathbb{P}^1$ of degree g whose restriction to every other component of $\mathbb{R}C$ has odd topological degree is separating. Such a morphism satisfies $d(f) = (2, 1, \dots, 1)$ after placing C' first. Thus C is not of special type if and only if, for every component C', there exists a nonseparating morphism of degree g with odd degree on all other real components and with $f(C') \neq \mathbb{RP}^1$. Its topological degree is $(0, 1, \dots, 1)$ after the same ordering.

The notion of special type introduced above gives another partition of the Teichmüller space:

$$\mathrm{Teich}_g = \mathrm{Teich}_{g,s} \cup \mathrm{Teich}_{g,ns}, \tag{‡}$$

where $\mathrm{Teich}_{g,s}$ and $\mathrm{Teich}_{g,ns}$ denote the subspaces of curves of special and non-special type, respectively.

In [Cop14], Coppens studies interactions between the partitions (†) and (‡). In particular, he proves the following statements.

(1) A separating $(M-2)$-curve that is not of special type has separating gonality $g-1$. Consequently, $\mathrm{Teich}_{g,g} \subseteq \mathrm{Teich}_{g,s}$.
(2) For every $g \geqslant 3$, there exist curves of special type with separating gonality $g-1$, as well as curves of special type with separating gonality g.
(3) If C is of special type and $\mathrm{sepgon}(C) = g-1$, then every separating pencil of degree $g-1$ on C is a theta characteristic.
(4) The locus $\mathrm{Teich}_{g,ns}$ is open. For $g \geqslant 4$, we have $\overline{\mathrm{Teich}_{g,ns}} = \mathrm{Teich}_{g,g-1}$.

Example 2.8. The separating gonality and type of low-genus separating $(M-2)$-curves whose separating semigroups are described in [Ore20, Ore25, MO25] are listed in Table 1.

g	**Model**	**Ovals**	$\mathrm{sepgon}(C)$	**Type**	$\mathrm{Sep}(C)$
3	Plane quartic	2	$3 = g$	Special	$(2,1) + \mathbb{N}_0^2$
3	Hyperelliptic	–	$2 = g-1$	Special	$\{(1,1)\} \cup \left(\mathbb{N}_{\geqslant 2} \times \mathbb{N}_{\geqslant 2}\right)$
4	Ellipsoid	3	$4 = g$	Special	$(2,1,1) + \mathbb{N}_0^3$
4	Quadratic cone	0	$3 = g-1$	Special	$\{(1,1,1)\} \cup \left((2,1,1) + \mathbb{N}_0^3\right)$
4	Quadratic cone	2	$4 = g$	Special	$(2,1,1) + \mathbb{N}_0^3$
4	Hyperboloid	0	$3 = g-1$	Non-special	$\mathbb{N}^3$
4	Hyperboloid	2	$4 = g$	Special	$(2,1,1) + \mathbb{N}_0^3$
6	Plane quintic	4	$6 = g$	Special	$(2,1,1,1,1) + \mathbb{N}_0^5$

Table 1. Separating gonality, special type, and separating semigroups of some separating $(M-2)$-curves of low genus.

Recall that if C is a real algebraic curve on a real algebraic surface S, then a connected component $C' \subset \mathbb{R}C$ is called an *oval* if it bounds a smooth disk in $\mathbb{R}S$.

Plane quartics and quintics are considered in $\mathbb{P}^2$. For $g = 4$, the column Model indicates the real quadric containing the canonical image of C in $\mathbb{P}^3$. Hyperelliptic curves are considered as

abstract curves, without a fixed embedding. Except in the hyperelliptic case, we number the real components so that $d_1(f) = 2$ for some separating morphism f of degree g.

Special type is not discussed in [Ore20, Ore25, MO25]. However, the types listed in the table can be easily deduced from the descriptions of the curves in these papers and Coppens's results [Cop14]. Therefore, we omit the proofs.

§2.3. A bit of Brill–Noether theory. In this subsection, C denotes a smooth projective irreducible curve (not necessarily real).

For integers d and $r \geqslant 0$, we define the Brill–Noether locus

$$W^r_d(C) := \{[D] \in \operatorname{Pic}^d(C) \mid h^0(D) \geqslant r+1\}.$$

If C is real, the locus $W^r_d(C)$ is invariant under the induced antiholomorphic involution on $\operatorname{Pic}^d(C)$ and is therefore defined over $\mathbb{R}$. We denote its real locus by $\mathbb{R}W^r_d(C)$.

We will need the following result from Brill–Noether theory.

Theorem 2.9 (Martens [Mar67])**.** *Let C be a smooth projective curve of genus g. Suppose that $0 < 2r \leqslant d < g$. Then $\dim W^r_d(C) \leqslant d - 2r$. Moreover, equality holds if and only if C is hyperelliptic.*

We will use Martens' bound to estimate the dimension of the complement of the locus of complete base-point-free pencils. We believe that the following technical statement is known (e.g. part of it is stated in [CH11, p. 16]), but we include a complete proof for the reader's convenience.

Proposition 2.10. *Let C be a smooth projective curve and let d be a positive integer. Define*

$$W^{1,\circ}_d(C) = \left\{[D] \in W^1_d(C) \;\middle|\; h^0(D) = 2,\ |D| \text{ is base-point-free}\right\}.$$

Then $W^{1,\circ}_d(C)$ is Zariski open in $W^1_d(C)$. If, in addition, C is non-hyperelliptic of genus g and $3 \leqslant d \leqslant g$, then

$$\dim\big(W^1_d(C) \setminus W^{1,\circ}_d(C)\big) \leqslant d-3.$$

Proof. We claim that

$$W^1_d(C) \setminus W^{1,\circ}_d(C) = W^1_{d-1}(C) + C,$$

where the sum is taken in $\operatorname{Pic}^d(C)$. Indeed, if $[D] \in W^1_d(C) \setminus W^{1,\circ}_d(C)$, then either $|D|$ has a base point p, in which case $h^0(D-p) = h^0(D) \geqslant 2$, or $h^0(D) \geqslant 3$, in which case $h^0(D-p) \geqslant 2$ for every $p \in C$. Thus $[D] \in W^1_{d-1}(C) + C$. Conversely, suppose that $D \sim E + p$ for some $[E] \in W^1_{d-1}(C)$ and $p \in C$. If $h^0(D) = h^0(E)$, then p is a base point of $|D|$; otherwise $h^0(D) > h^0(E) \geqslant 2$, and hence $h^0(D) \geqslant 3$. In either case, $[D] \notin W^{1,\circ}_d(C)$.

The locus $W^1_{d-1}(C) + C$ is Zariski closed in $\operatorname{Pic}^d(C)$, so $W^{1,\circ}_d(C)$ is Zariski open in $W^1_d(C)$.

Finally, suppose that C is non-hyperelliptic and $3 \leqslant d \leqslant g$. Martens' theorem applied to $W^1_{d-1}(C)$ gives

$$\dim W^1_{d-1}(C) \leqslant d-4.$$

Consequently,

$$\dim\big(W^1_d(C) \setminus W^{1,\circ}_d(C)\big) \leqslant \dim W^1_{d-1}(C) + 1 \leqslant d-3.$$

□

Remark 2.11. As in [CH11, p. 15] and in [Cop14], we will use the standard fact that the topological degree vector $\delta(f_{|D|})$ is locally constant on the locus $\mathbb{R}W^{1,\circ}_d(C)$ of complete base-point-free real pencils.

§3. Proof of the main results

§3.1. Curves of special type.

Theorem 3.1. *Let C be a separating $(M-2)$-curve of genus $g \geqslant 4$.*

(a) *Assume that* $\operatorname{sepgon}(C) = g$ *and let f be a separating morphism of degree g. Number the connected components of $\mathbb{R}C$ so that $d_1(f) = 2$. Then*

$$(2, 1, \ldots, 1) + \mathbb{N}_0^{g-1} \subseteq \operatorname{Sep}(C).$$

(b) *Assume that* $\operatorname{sepgon}(C) = g-1$ *and C is of special type with a special component C_1. Then*

$$\{(1, \ldots, 1)\} \cup \big((2, 1, \ldots, 1) + \mathbb{N}_0^{g-1}\big) \subseteq \operatorname{Sep}(C).$$

Proof. **(a)** Let $D = (f)_0$. By the Riemann–Roch theorem, $h^0(D) = h^0(K_C - D) + 1$, so $h^0(K_C - D) \geqslant 1$. Choose a real divisor $E \in |K_C - D|$. The canonical class has even degree on every connected component of $\mathbb{R}C$ (see e.g. [Hui03, Proposition 2.1] or [Mag26, Lemma 3]), whereas D has even degree on C_1 and odd degree on every other connected component. Hence E has odd degree on $C_2, \ldots, C_{g-1}$. Since $\deg E = g-2$, we have $E = q_2 + \ldots + q_{g-1}$ where $q_i \in C_i$ for $i = 2, \ldots, g-1$. By Proposition 2.1 we have $h^0(E) = 1$ and therefore $h^1(D) = 1$.

Thus the claim follows from part (a) of Theorem 2.4.

(b) Since the number of connected components of the real locus of a separating hyperelliptic curve of genus g can be 1, 2, or $g+1$ (see [GH81, Prop. 6.3]), a real separating $(M-2)$-curve cannot be hyperelliptic for $g \geqslant 4$.

Since $\operatorname{sepgon}(C) = g-1$, we first have $(1, \ldots, 1) \in \operatorname{Sep}(C)$. It remains to prove that $(2, 1, \ldots, 1) + \mathbb{N}_0^{g-1} \subseteq \operatorname{Sep}(C)$. Since C is of special type with special component C_1, every real morphism of degree g with odd topological degree on $C_2, \ldots, C_{g-1}$ is separating. Therefore, to construct a separating morphism with degree vector $(2, 1, \ldots, 1)$, it suffices to find a complete base-point-free pencil of the form $|K_C - Q|$ with

$$Q = q_2 + \cdots + q_{g-1}, \qquad q_i \in C_i.$$

Such divisors Q are parametrized by the $(g-2)$-dimensional real torus $\mathscr{T} = C_2 \times \cdots \times C_{g-1}$. We identify this torus with the corresponding subspace of $\mathbb{R}\operatorname{Sym}^{g-2}(C)$ and consider the restriction of the Abel–Jacobi map:

$$\varphi \colon \mathscr{T} \longrightarrow \mathbb{R}\operatorname{Pic}^{g-2}(C), \qquad (q_2, \ldots, q_{g-1}) \longmapsto [Q],$$

where $Q = q_2 + \cdots + q_{g-1}$. Consider also the isomorphism

$$\iota_K \colon \mathbb{R}\operatorname{Pic}^{g-2}(C) \longrightarrow \mathbb{R}\operatorname{Pic}^{g}(C), \qquad [D] \longmapsto [K_C - D].$$

Set $\Phi = \iota_K \circ \varphi$ and $\Lambda = \operatorname{Im}(\Phi)$. For every such Q, Proposition 2.1 gives $h^0(Q) = 1$, and the Riemann–Roch theorem then gives $h^0(K_C - Q) = 2$. Therefore, $\Lambda \subset \mathbb{R}W_g^1(C) \setminus \mathbb{R}W_g^2(C)$.

We claim that Φ is a (smooth) embedding. Indeed, the differential of the Abel–Jacobi map at Q has rank $g - h^0(K_C - Q) = g-2$ and thus it is injective. Its restriction to the tangent space of $\mathscr{T}$ is therefore also injective. Since ι_K is an isomorphism, it follows that $d_Q\Phi$ is injective and thus Φ is a real-analytic immersion. Also, Φ is injective. Indeed, if $\Phi(Q) = \Phi(Q')$, then $Q \sim Q'$, and as $\dim |Q| = 0$, it follows that $Q = Q'$. Therefore, Φ is an injective immersion, and since $\mathscr{T}$ is compact, the claim is proven. Consequently, Λ is a smooth real submanifold of $\mathbb{R}\operatorname{Pic}^{g}(C)$ of dimension $g-2$.

On the other hand, Proposition 2.10 gives

$$\dim_{\mathbb{R}}\big(\mathbb{R}W_g^1(C) \setminus \mathbb{R}W_g^{1,\circ}(C)\big) \leqslant g-3.$$

Therefore, Λ is not contained in this locus. Hence we can choose Q such that $|K_C - Q|$ defines a separating morphism f with $d(f) = (2, 1, \dots, 1)$. Let $D = (f)_0$. Since $D \sim K_C - Q$, we have $h^1(D) = h^0(Q) = 1$. Thus part (a) of Theorem 2.4 gives that

$$(2, 1, \dots, 1) + \mathbb{N}_0^{g-1} \subseteq \mathrm{Sep}(C).$$

□

§3.2. Curves not of special type. In this subsection, C denotes a separating $(M-2)$-curve of genus $g \geqslant 5$ that is not of special type.

Coppens [Cop14] proved that every separating $(M-2)$-curve that is not of special type has separating gonality $g-1$. By part (b) of Theorem 2.4, to prove that $\mathrm{Sep}(C) = \mathbb{N}^{g-1}$, it therefore suffices to find a separating divisor of degree $g-1$ that is not a theta characteristic.

However, such a divisor does not always exist: Coppens also proved that if the curve is of special type and has separating gonality $g-1$, then every separating pencil of degree $g-1$ is a theta characteristic. Here we prove the converse (Theorem 3.5): if C is not of special type, then it admits a separating pencil of degree $g-1$ which is not a theta characteristic.

To prove Theorem 3.5, we will compare topological degrees on C_1 of a separating and a nonseparating pencil of degree g, as in Coppens's proof of Theorem 4.4 in [Cop14]. As mentioned in the proof of Theorem 3.1, every real divisor D of degree g with $h^0(D) \geqslant 2$ and odd degree on $C_2, \dots, C_{g-1}$ is linearly equivalent to $K_C - Q$ for some

$$Q = q_2 + \cdots + q_{g-1}, \qquad q_i \in C_i.$$

We use the same notation $\mathscr{T} = C_2 \times \cdots \times C_{g-1}$ for the torus parametrizing such divisors Q. Recall that $h^0(K_C - Q) = 2$ for every $Q \in \mathscr{T}$.

Outline of the proof of Theorem 3.5.

Step 0. Suppose for a contradiction that every separating pencil of degree $g-1$ on C is a theta characteristic.

Step 1. In Lemma 3.3, we prove that there exists a separating pencil $|D_1|$ of degree g. We number the connected components of $\mathbb{R}C$ so that $d(f_{|D_1|}) = (2, 1, \dots, 1)$.

Step 2. Since C_1 is not special, there exists a real morphism $h\colon C \to \mathbb{P}^1$ of degree g such that

$$h(C_1) \subsetneq \mathbb{RP}^1, \qquad \delta_i(h) \text{ is odd for } i = 2, \dots, g-1.$$

In particular, $\deg(h|_{C_1}) = 0$. Put $D_2 = (h)_0$.

Step 3. By the preceding discussion, we can write

$$|D_1| = |K_C - Q_+|, \qquad |D_2| = |K_C - Q_-|, \qquad Q_+, Q_- \in \mathscr{T}.$$

Step 4. Consider the locus

$$\mathscr{B}_1 = \{Q \in \mathscr{T} \mid |K_C - Q| \text{ has a base point on } C_1\}.$$

Under the assumption of Step 0, we show in Lemma 3.4 that $\mathscr{B}_1$ is contained in a finite union of embedded circles. Since $\dim \mathscr{T} = g - 2 \geqslant 3$, the complement $\mathscr{T} \setminus \mathscr{B}_1$ is connected. Both pencils $|D_1|$ and $|D_2|$ are base-point-free, so $Q_+, Q_- \in \mathscr{T} \setminus \mathscr{B}_1$.

Step 5. For every $Q \in \mathscr{T} \setminus \mathscr{B}_1$, restriction of the pencil $|K_C - Q|$ to C_1 defines a smooth map $\varphi_Q\colon C_1 \to \mathbb{RP}^1$. Locally, these maps vary continuously with Q, so the condition $\deg(\varphi_Q) = 0$ is locally constant on $\mathscr{T} \setminus \mathscr{B}_1$. However,

$$\deg(\varphi_{Q_+}) \neq 0, \qquad \deg(\varphi_{Q_-}) = 0,$$

contradicting the connectedness of $\mathscr{T} \setminus \mathscr{B}_1$.

The following lemma will be used in the proof of Lemma 3.3.

Lemma 3.2. *Assume that every separating pencil of degree $g-1$ on C is a theta characteristic. Let $f\colon C \to \mathbb{P}^1$ be a separating morphism of degree $g-1$. Then there exist two distinct real fibers P, Q of f such that the only separating divisors $D \leqslant P+Q$ of degree $g-1$ are P and Q.*

Proof. Set $P = (f)_0 = p_1 + \cdots + p_{g-1}$, where $p_i \in C_i$. By Proposition 2.3, we have $h^0(P) = 2$, so P is a theta characteristic by assumption. Since theta characteristics form a finite subset of $\mathrm{Pic}^{g-1}(C)$, there exists a neighborhood $U \ni [P]$ which does not contain any other theta characteristics.

Choose a local coordinate t near $0 \in \mathbb{RP}^1$. For sufficiently small $\varepsilon > 0$, write

$$f^{-1}(t) = p_1(t) + \cdots + p_{g-1}(t),$$

where $p_i(t)\colon (-\varepsilon, \varepsilon) \to \mathbb{R}C$ are smooth paths such that $p_i(0) = p_i$. Since $p_i(t) \to p_i$ as $t \to 0$ for every i, we can choose a sufficiently small $t_0 \neq 0$ so that $[D] \in U$ for every effective divisor $D \leqslant P + f^{-1}(t_0)$ containing exactly one point on each connected component of $\mathbb{R}C$. Set $Q = f^{-1}(t_0)$.

Let $D \leqslant P+Q$ be a separating divisor of degree $g-1$. Then D contains exactly one point on each real component and is a theta characteristic by assumption. By the choice of U, we have $D \in |P|$. Since $h^0(P) = 2$, D is a fiber of f, so $D = P$ or $D = Q$. □

Lemma 3.3. *Assume that* $\mathrm{sepgon}(C) = g-1$ *and that every separating pencil of degree $g-1$ on C is a theta characteristic. Then there exists a separating divisor of degree g.*

Proof. Let $f\colon C \to \mathbb{P}^1$ be a separating morphism of degree $g-1$. By assumption, $(f)_0$ is a theta characteristic. Choose two real fibers

$$P = p_1 + \cdots + p_{g-1}, \qquad Q = q_1 + \cdots + q_{g-1}$$

as in Lemma 3.2. Then $P + Q \sim K_C$.

Below, we use the same idea as in the proof of Theorem 1 in [Mag26]. Let $\varphi_n\colon \mathrm{Sym}^n(C) \to \mathrm{Pic}^n(C)$ denote the Abel–Jacobi map, and fix a complex orientation on $\mathbb{R}C$. By [Mag26, Lemma 1], there exist tangent vectors $v_i \in T_{p_i}(\mathbb{R}C)$ and $u_i \in T_{q_i}(\mathbb{R}C)$, all positive with respect to the chosen orientation, such that

$$(v_1, \ldots, v_{g-1}) \in \mathrm{Ker}(d_P\varphi_{g-1}), \qquad (u_1, \ldots, u_{g-1}) \in \mathrm{Ker}(d_Q\varphi_{g-1}).$$

Using the natural inclusion

$$\mathrm{Ker}(d_P\varphi_{g-1}) \oplus \mathrm{Ker}(d_Q\varphi_{g-1}) \hookrightarrow \mathrm{Ker}(d_{P+Q}\varphi_{2g-2}),$$

we may consider the vectors

$$v = (v_1, \ldots, v_{g-1}, 0, \ldots, 0), \qquad u = (0, \ldots, 0, u_1, \ldots, u_{g-1})$$

as elements of $\mathrm{Ker}(d_{P+Q}\varphi_{2g-2})$. Since $g \geqslant 4$, we have

$$\dim \mathrm{Ker}(d_{P+Q}\varphi_{2g-2}) = \dim |P+Q| = \dim |K_C| = g-1 \geqslant 3.$$

Therefore, we may choose a real vector $w \in \mathrm{Ker}(d_{P+Q}\varphi_{2g-2})$ such that $w \notin \mathrm{span}_{\mathbb{R}}(v, u)$. Since all the tangent spaces $T_{p_i}(\mathbb{R}C)$ and $T_{q_i}(\mathbb{R}C)$ are one-dimensional, we may write

$$w = (\alpha_1 v_1, \ldots, \alpha_{g-1} v_{g-1}, \beta_1 u_1, \ldots, \beta_{g-1} u_{g-1}), \qquad \alpha_i, \beta_i \in \mathbb{R}.$$

Denote $\alpha := \min\limits_i(\alpha_i)$, $\beta := \min\limits_i(\beta_i)$ and consider the vector

$$\widetilde{w} = w - \alpha v - \beta u.$$

Then $\widetilde{w}$ is a nonzero vector in $\operatorname{Ker}(d_{P+Q}\varphi_{2g-2})$, and all its components are nonnegative (i.e. zero or positive with respect to the chosen complex orientation). Let $I = \{i = 1, \dots, 2g-2 \mid \widetilde{w}_i \neq 0\}$, and consider the divisor

$$D = \sum_{\substack{i \in I \\ i \leqslant g-1}} p_i + \sum_{\substack{i \in I \\ i \geqslant g}} q_{i-g+1}.$$

By [Mag26, Lemma 1], D is separating. Moreover, D contains neither P nor Q, since $\widetilde{w}$ has a zero component in each of the two blocks. Since $D \leqslant P + Q \sim K_C$, the divisor D is special. By [Mag26, Corollary 2], there exists a separating divisor $D' \leqslant D$ with $\deg D' \leqslant g$. Since $\operatorname{sepgon}(C) = g-1$, we have

$$g - 1 \leqslant \deg D' \leqslant g.$$

If $\deg D' = g-1$, Lemma 3.2 gives $D' = P$ or $D' = Q$. This is impossible because $D' \leqslant D$ and D contains neither P nor Q. Thus $\deg D' = g$, as required. □

Lemma 3.4. *Assume that* $\operatorname{sepgon}(C) = g-1$ *and that every separating pencil of degree* $g-1$ *on* C *is a theta characteristic. Consider the locus*

$$\mathscr{B}_1 = \{Q \in \mathscr{T} \mid |K_C - Q| \text{ has a base point on } C_1\}.$$

Then $\mathscr{T} \setminus \mathscr{B}_1$ *is connected.*

Proof. Let $Q \in \mathscr{B}_1$, and let $p \in C_1$ be a base point of $|K_C - Q|$. Then $h^0(K_C - Q - p) = h^0(K_C - Q) = 2$. Thus, by the Riemann–Roch theorem, $h^0(Q + p) = 2$.

Removing any point from $Q+p$ leaves $g-2$ points on distinct components, so Proposition 2.1 shows that $|Q + p|$ is base-point-free. By part (b) of Proposition 2.3, $|Q + p|$ is a separating pencil of degree $g - 1$. By assumption, $Q + p \in |E|$ for some theta characteristic $[E]$. Set $f = f_{|E|}$. Since f is a separating morphism of degree $g - 1$, we have $d_i(f) = 1$ for every i. Hence each restriction

$$f|_{C_i} \colon C_i \longrightarrow \mathbb{RP}^1$$

is a diffeomorphism. Denote its inverse by $q_i \colon \mathbb{RP}^1 \to C_i$. Thus, for every $t \in \mathbb{RP}^1$,

$$f^{-1}(t) = q_1(t) + \cdots + q_{g-1}(t).$$

Therefore, the map

$$\gamma_{|E|} \colon \mathbb{RP}^1 \longrightarrow \mathscr{T}, \qquad t \longmapsto \big(q_2(t), \dots, q_{g-1}(t)\big),$$

is an embedding, and its image is a circle in $\mathscr{T}$ containing Q. Since there are only finitely many theta characteristics on C, it follows that $\mathscr{B}_1$ is contained in a finite union of embedded circles in $\mathscr{T}$. As $\dim \mathscr{T} = g - 2 \geqslant 3$, such a subset cannot disconnect $\mathscr{T}$. □

Theorem 3.5. *Suppose that* $\operatorname{sepgon}(C) = g-1$, *where* $g \geqslant 5$, *and that* C *is not of special type. Then there exists a separating pencil of degree* $g-1$ *that is not a theta characteristic.*

Proof. Suppose for a contradiction that every separating pencil of degree $g-1$ on C is a theta characteristic.

By Lemma 3.3, there exists a separating morphism $f \colon C \to \mathbb{P}^1$ of degree g. Number the connected components of $\mathbb{R}C$ so that $d(f) = (2, 1, \dots, 1)$. With this numbering, we use the notation $\mathscr{T}$ and $\mathscr{B}_1$ introduced above.

For $Q \in \mathscr{T} \setminus \mathscr{B}_1$, the pencil $|K_C - Q|$ has no base points on C_1. Hence its restriction to C_1 defines a smooth map

$$\varphi_Q \colon C_1 \approx S^1 \longrightarrow S^1 \approx \mathbb{RP}^1.$$

Although its topological degree is defined only up to sign, its vanishing is well defined. The condition $\deg(\varphi_Q) = 0$ is locally constant on $\mathscr{T} \setminus \mathscr{B}_1$ (cf. Remark 2.11). Since $\mathscr{T} \setminus \mathscr{B}_1$ is

connected by Lemma 3.4, this condition must hold either everywhere or nowhere. To obtain a contradiction, it therefore suffices to find $Q_+, Q_- \in \mathscr{T} \setminus \mathscr{B}_1$ such that

$$\deg(\varphi_{Q_+}) \neq 0, \qquad \deg(\varphi_{Q_-}) = 0.$$

Construction of Q_+. Put $D = (f)_0$. As explained above, we can write $D \sim K_C - Q_+$ with $Q_+ \in \mathscr{T}$. The pencil $|K_C - Q_+|$ defines f, so $Q_+ \notin \mathscr{B}_1$ and $\deg(\varphi_{Q_+}) \neq 0$.

Construction of Q_-. Since C_1 is not special, there exists a real morphism $h\colon C \to \mathbb{P}^1$ of degree g such that $h(C_1) \subsetneq \mathbb{RP}^1$ and $\delta(h) = (0, 1, \ldots, 1)$. Put $P = (h)_0$. Similarly, we can write $P \sim K_C - Q_-$ with $Q_- \in \mathscr{T}$. Thus $|P| = |K_C - Q_-|$ is precisely the pencil defining h. This pencil is base-point-free, so $Q_- \notin \mathscr{B}_1$. Since $\delta_1(h) = 0$, we have $\deg(\varphi_{Q_-}) = 0$. This gives the required contradiction.

□

Combining Theorem 3.5 with [Cop14, Proposition 4.1], we obtain the following characterization of curves of special type with separating gonality $g-1$.

Corollary 3.6. *Let C be a separating $(M-2)$-curve of genus $g \geqslant 5$ with $\operatorname{sepgon}(C) = g-1$. Then C is of special type if and only if every separating pencil of degree $g-1$ on C is a theta characteristic.*

Theorem 3.7. *Let C be a separating $(M-2)$-curve of genus g. Assume that $g \geqslant 5$ and C is not of special type. Then*

$$\operatorname{Sep}(C) = \mathbb{N}^{g-1}.$$

Proof. We apply part (b) of Theorem 2.4. The curve C is non-hyperelliptic (see the proof of part (b) of Theorem 3.1).

By Theorem 3.5, there exists a separating pencil $|Q|$ of degree $g-1$ which is not a theta characteristic. We have $d(Q) = (1, \ldots, 1)$, and Proposition 2.3 gives $h^0(Q) = 2$. Part (b) of Theorem 2.4 therefore shows that $(1, \ldots, 1)$ is extendable and thus $\operatorname{Sep}(C) = \mathbb{N}^{g-1}$. □

As mentioned in §2.2, Coppens [Cop14, Lemma 2.6 and Corollary 4.2] showed that $\operatorname{Teich}_{g,ns}$ is a non-empty open subset of Teich_g. Combining this result with Theorem 3.7, we obtain the following corollary.

Corollary 3.8. *Let $g \geqslant 5$. On a non-empty open subset of Teich_g, the separating semigroup equals $\mathbb{N}^{g-1}$.*

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